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Introduction:

Scientists widely accept climate change as “the defining issue of our time”(UN); it is one of the most complex collective action cases philosophers have ever faced. Greenhouse gas(GHG) emissions from human activity create a greenhouse effect that will rapidly change the climate. Several vital human activities, including farming, transportation, and electricity production, burn fossil fuels, which releases GHGs. Industries like energy production and transportation have been crucial for improving global living standards over the past few centuries. However, climate change threats are rapidly outpacing the benefits we get from using fossil fuels that burn GHGs(Ackerman). Climate change is dangerous because it increases both the frequency and intensity of “extreme weather”(Carrington) events like hurricanes or floods and the pace of this increase is expanding exponentially. In the Journal of Bioscience, “11,000 scientists” leading climate scientists from across the planet warned that we are currently in a “climate emergency,” that it is “more severe than anticipated.” If countries do not take drastic action to mitigate the coming disaster, the very “fate of humanity”(Carrington) is at risk. Besides other animals on the planet, humans could go extinct if the world does not address climate change(Carrington). Steven Hawking, who is widely considered to be one of the most brilliant scientists of the 20th and 21st centuries, warned that:

Climate change “could push the Earth over the brink, to become like Venus, with a temperature of two hundred and fifty degrees, and raining sulphuric acid”(Kharpal).

The paper argues for three main premises to outline the moral obligations climate change brings up. The first argument illustrates how burning fossil fuels creates or worsens extreme

weather events and that because this is human-made, it leads to human rights violations. The second explains why countries as entities bear responsibility for all of their emissions, past, and present. The third describes an egalitarian argument for why countries have obligations to protect future generations' rights and why governments have some leeway in implementing climate change mitigation tactics, given the apocalyptic consequences that climate change presents.

Given these three points, the paper asserts that, because of the dire warning from climate scientists in the *Journal of Bioscience*, any significant reduction in the GHGs a country emits would allot humanity more crucial time to move away from fossil fuels fully. These apocalyptic visions about the future, coupled with how climate change currently threatens people's human rights, mean that all countries have an obligation to take drastic action to reduce their emissions as much as possible.

What does the thesis mean by drastic action? Countries should do whatever they can to reduce their GHG emissions, as long as it does not violate their citizens' rights in the present. For example, there is evidence that the US can convert to 100% renewables for “4.5 trillion” according to “the energy research firm Wood Mackenzie”(Yale). Further, nuclear energy seems to be an option for wealthier countries, France gets “75% of its electricity from nuclear energy”(WNA), and now France has a “very low cost”(WNA) of electricity generation. Solar panels recently became the cheapest way to produce energy, which increases the options for developing nations when expanding their grids(Ritchie). Countries like France also have plans to ban all gas-powered vehicles by 2040(BBC). Even America's move from coal to natural gas has significantly reduced its emissions(UCSUSA). Massive changes like these significantly reduce total GHG emissions and allot humanity more time to move away from fossil fuels fully. The paper does not explicitly advocate for any of these actions, nor that we return to pre-industrial

living and stop using electricity. Nevertheless, we need to demonstrate that scientists continuously innovate and find new ways to produce energy and that fossil fuels are not the only option to produce energy. There is still hope. The paper advocates that we take all new data seriously and allow skilled engineers/legislators to tackle the issue in the best way possible.

Premise One:

The first premise is that by emitting GHGs, countries violate people's human rights in the present and near-future who are being adversely affected by extreme weather events. For the sake of this argument, the paper will utilize the UN's groundbreaking Universal Declaration of Human Rights(UDHR) for its definition of human rights. The paper will assume that it is self-evident that these rights should not be infringed upon because almost every country has agreed to uphold them, and the paper focuses on the obligations of countries. Climate scientists have suggested the increase in the severity of "category 4 and 5"(Berardelli) hurricanes devastating countries from the US to the Philippines is due to climate change, specifically warming ocean temperatures. The Union of Concerned Scientists has compelling evidence showing a linear relationship between the general increase in wildfires and the average increase in temperatures(UCSUSA). There has also been significant flooding in the gulf area; the US is relocating "entire towns"(Bagri) due to this flooding costing millions of dollars. There has also been horrible climate change-related flooding in countries like Pakistan, where:

"in 2010 high Atlantic Ocean temperatures contributed to the intense monsoons that led to catastrophic flooding in Pakistan, causing over 2000 deaths and displacing 20 million people, 85% of whom were women and children"(Cuomo 693).

One could argue that because climate change is human-made and causes or worsens all of these extreme weather conditions, the countries who contribute to it violate Article 25 of the UDHR; they directly infringed upon people's housing rights. These examples should clarify why billions of people driving to work every day can violate human rights. People have to drive to get to work and participate in their economies, and burning fossil fuels is not as clear-cut a case as human rights violations like murder, human trafficking, or genocide. Regardless, climate change has real and severe consequences for people every day. Emitting GHGs on a large scale is not a simple, benign, or neutral act; it has led to people's deaths, displacements, and malnourishment. Therefore, burning fossil fuels violates people's fundamental rights to "life, liberty and security"(UN) as described in Article 3 of the UDHR. Further, all of these examples from the past 20 years are just the tip of the iceberg; there are dozens of other examples of disasters that scientists have been able to attribute to climate change.

As time goes by, scientists are getting a lot better at attributing specific natural disasters to climate change. There is a new field in climate science called "extreme event attribution"(Harvey), which uses data to predict whether climate change causes or worsens a specific event. This field is growing daily, and the precision/accuracy of the predictions is increasing. However, even now, with some events, scientists can predict that climate change made a flood much worse with a "confidence level [of about] 90%"(Harvey). This new data is a real game-changer in terms of climate justice, it allows us to understand the kind of damage climate change is doing to a significantly higher degree of accuracy. It is much easier to claim that climate change violates human rights with high quality empirical data.

If emitting GHGs violates people's rights by causing these destructive weather events, countries should emit as few GHGs as possible. This argument is key to proving the thesis. However,

premise 1 runs into a significant epistemic issue - cluelessness. James Lenman explained his cluelessness objection to consequentialism in his paper *Consequentialism and Cluelessness*. Lenman correctly points out that it is “impossible to know the future”(Lenman 343), and thus when partaking in an action, it is not clear what the consequences will be. The nature of reality means that there is simply no way of knowing how one’s actions will affect the future. This concept pertains to climate justice, and this paper because it highlights that we cannot see how a specific agent’s greenhouse emissions will affect the future.

The CO₂ from a car ride to the grocery store could all get absorbed by nearby trees, while the CO₂ from the ride back could warm the earth by a few more nano degrees celsius. This cluelessness is complicated further because the relationship between emissions and a warming climate “is not linear”(Miller 131). The nature of fluid mechanics in the atmosphere is incredibly complicated, which is why it has been challenging to attribute things like hurricanes to climate change accurately. Consequently, muddying the thesis's ability to say that climate change harms human rights because it is difficult to know if certain events are caused by it and to what extent.

The new field of extreme event attribution has made significant progress in the realm of empirical uncertainty. One can no longer say that we cannot attribute a specific calamity to climate change. Further, aspects of cluelessness do not work well when dealing with states (and not individuals) as actors or when looking at aggregate trends from the past. Scientists have empirical data on each state's total emissions, and one can attribute this to a percentage of how much the earth warmed(Ritchie). Moreover, many scientists would probably disagree with the sentiment that a drive to the movie theatres does not cause climate change; it 100% emits GHGs, and GHGs absorb heat on a molecular level. The problem is that cluelessness is not so easy to

sidestep. Even with the ability to attribute a specific event to climate change (even if we somehow were 100% certain about the science), we cannot know the future.

Truly embracing cluelessness leads to arguments along the lines of - maybe a scientist will create some new cheap form of energy that is very easy for everyone to switch to tomorrow. Maybe an asteroid will strike the earth in 2100, so it will be destroyed either way, and climate change will not make anything worse in the aggregate. Even this example may not be sufficiently fantastical. Cluelessness could even entertain the thought that it may be better for the human race to go extinct; a new species of intelligent dolphins with much better morals will arise in the aftermath and use the earth much better than humans ever could. The point is that we do not know. If we do not know what may happen in the future, Premise One and the thesis are false. Even if we have successfully argued against most of the more scientific aspects of cluelessness in the context of climate change, and the cluelessness argument now only rests on the idea we are fundamentally ignorant of the future, we are in an epistemic rut. We will need an equally powerful epistemic objection to cluelessness to get ourselves out of this rut.

Avram Hiller brilliantly tackles the cluelessness argument in his paper *Climate Change and Individual Responsibility*. He explains that even if we do not know the results of a particular action, there is sufficient mathematical and scientific knowledge that allows us to calculate the probability that an act leads to dangerous climatic events. So, we know the expected value of doing a specific action; therefore, we can understand the “expected harm” of burning fossil fuels and that it is “nonzero” (Hiller 353). Extreme event attribution further bolsters this point. This new field gives climate scientists and philosophers the ability to compute expected harms to a significantly higher degree of certainty when dealing with large amounts of pollution from countries. Because we understand this expected harm, Hiller suggests that there is a *prima facie*

duty not to commit actions that cause expected damage, even if we do not 100% know the result of said action. A *prima facie* duty is one that applies to a situation by default. This argument is the exact kind of powerful epistemic rebuttal we need to save Premise One and the thesis. Morally, it is convincing; the idea of something being *prima facie* wrong is not a new one in philosophy.

In the same way that something like reckless/drunk driving is *prima facie* wrong, even if the car does not hit anyone, a country that performs an action that worsens climate change is also *prima facie* wrong. Hiller argues about these duties regarding individuals, but the paper adapts his idea for countries and collectives. The probability that burning fossil fuels will lead to human rights abuses is high enough for the action to be wrong by default. This deontological conception is in line with the “law of torts”(Moellendorf 135), which defines the responsibility for accidents as having knowingly and voluntarily taken a risky action that leads to bad outcomes. Hiller’s response still does not mean that we know the future, but we know the probability that a certain amount of GHG emissions will cause or worsen extreme weather events. In terms of the more outlandish scenarios cluelessness produces, it does not seem sufficiently convincing that we should overlook these expected harms because of some magical solution to the issue of producing energy(or a potential race of super-dolphins). This line of reasoning makes it impossible to argue anything or to get anything done. Therefore, the paper will assume that the cluelessness objection has been sufficiently refuted by considering actions that lead to climate change as *prima facie* wrong, so premise one and the thesis are still valid. A country’s contribution to climate change is not wrong in the same sense that murder is wrong; it is *prima facie* wrong.

The idea of an action being wrong by default (and its empirical backing), as well as the dire warning from the Journal of Bioscience, show why countries have a moral obligation to emit as little as possible as the thesis states. The paper will hope to further cement this idea in the coming paragraphs.

Now that the paper has introduced the idea of expected harm, it should be more apparent why the thesis defines actors/polluters as countries/groups instead of individuals. Although one could argue that there is individual responsibility here, it seems better to think about the problem in terms of countries. Even if an individual drive to work has a negative expected utility, governments create laws and guidelines around how the cars should be built and which cars can be sold. Weak regulations around car emissions have a much greater and more calculable negative expected value than one person's drive to work. For example, if a country like the US theoretically banned all non-electric vehicles or banned beef consumption, it would reduce its emissions by a significant percentage. Whereas if a quarter, half, or even three-quarters of individuals decided to use electric vehicles and stop eating beef right now, it would reduce emissions, but not as much. Government regulations have significantly more sway in these situations; they make a much more significant dent in emissions. Due to the power to regulate, a country seems like it is a more morally significant target than individuals.

Hopefully, this section has shown why the emission of GHGs has moral significance, how climate change ruins people's lives, and why a country emitting a GHG is *prima facie* impermissible. We can now move onto why countries should bear the costs of their emissions.

Premise Two:

The second premise is that countries are responsible for the damage produced by all of their past and present emissions. Why is this? In defending premise one, the paper argues that GHG emissions are *prima facie* wrong when one knows there is a significant probability of said emissions leading to human rights abuses (so after 1990). So, why are countries responsible for their emissions before they knew climate change existed? The concept that one is responsible for one's actions, regardless of intent/knowledge, is not new. For example, in most countries, one cannot use ignorance as an excuse for committing a crime, whether it be going over the speed limit, littering, or an extreme case like murder, the legality of an action is not tied to whether one knew it was a crime.

However, driving to work in an economy based on fossil fuels is not the same as littering, so this analogy is insufficient to prove the point. The paper's main logic behind countries being responsible for their past emissions is the "Beneficiary responsibility" (Moellendorf 136) principle (or beneficiary pays principle BPP), which suggests that people who have benefitted from past emissions should pay for said emissions.

Countries that industrialized first such as Japan, the US, the UK, and France, have gained significant benefits from releasing GHGs. The energy they have been able to use post-industrialization has dramatically improved the material well-being of their citizens. In the 18th century, the ancestors of the people in developed nations like Japan or France were farmers making subsistence livings with incredibly high mortality rates. Now, the average person lives better than royalty did back then. Although, the first countries to industrialize are not the only ones seeing significant benefits from emitting GHGs. Since "1970" China has managed to lift

“hundreds of millions of”(Harrison) their people out of poverty (close to a billion), and they emitted “12.7%”(Ritchie) of total cumulative global emissions to do so.

Similarly, since 1990, experts estimate that India managed to lift around “271 million”(Mccarthy) out of poverty; India is now responsible for around “7%”(UCSUSA) of yearly global emissions. There are also similar stories from other recently developed countries. This data is not to say that this development has been bad, nor that there is a per-capita comparison between Chinese and American emissions. However, countries like China have demonstrably gained significant benefits from their government’s emissions. Further, the benefits that countries like the US and China have gained from their emissions have not come at 0 costs. Premise one went into considerable detail explaining the countless disasters caused or made worse by climate change. All GHG emissions, past and present, threaten people’s rights in the present. Countries like the UK and China have developed their countries significantly using GHGs, however “small island”(Page 163) nations like Fiji have demonstrably insignificant emissions, and they may soon be underwater because of said emissions(UCSUSA). It does not seem fair that people who have not contributed to nor benefited from climate change in any way will face the steepest consequences as a result of it. If we take these inequities into account, protecting people's rights in places like Fiji seems even more crucial.

However, the BPP has a significant epistemic issue that we must consider, the non-identity problem. The main point of the non-identity problem is that if one’s parents did one thing differently in the past, they would no longer exist; therefore, if people in the past had not polluted, no one in the present would exist. This argument means that past actions cannot make people in the future better or worse off because any change in the past changes the individuals who live in the future. Derek Parfit first brought this idea up in his writing *Reasons and Persons*.

When considering individuals as polluters, it is difficult to argue against the non-identity problem. As Caney puts it, we cannot tell an individual that they have benefitted or been harmed by past actions because “without industrialization,” they “would not exist”(Caney 758). So, how can we say that past emissions have hindered the lives of people in the present? Luckily, the non-identity problem does not fit as well when considering collectives; Caney explains that:

“whereas we cannot say that industrialization has bestowed (net)advantages on currently existing individuals that they would otherwise be without, we can say that industrialization has bestowed (net) advantages on currently existing countries (such as Britain) that they would otherwise be without. The collectivist response thus enables us to defend the BPP”(Caney 759).

This line of reasoning is why the thesis specifies that countries, not individuals, are responsible for past emissions. We can hardly say that the USA would not exist if it had switched more of its grid over to nuclear in the 80s and 90s like France. Similarly, we cannot say that Fiji would not exist if the UK had switched to solar energy in the past. Caney still thinks that attributing the moral responsibility to countries is inadequate because many people in the present had nothing to do with the decisions that lead to climate change “they were not consulted”(Caney 760). However, whether people were consulted is not a sufficient rebuttal to the BPP; these countries have benefitted significantly from these past actions. Countries' past and present emissions hinder millions of people in the present. Further, the continuation of business as usual emissions will, at the very least, hinder the lives of billions in the future(Carrington). At worst, they will lead to humanity's extinction(Carrington). The concept that democratic liberal

countries should take responsibility for their past actions is not a new one. Rawles explains that from a cosmopolitan point of view, "self-governing peoples" ought to take "responsibility for their policies"(Caney 759) in the past, as well as the present. The paper agrees with this cosmopolitan view; "liberal or decent"(Caney 759) countries need to take responsibility for their past actions.

At this point, we have argued for why countries are responsible for their past emissions. Now we must ask, what obligations do people in the present have to future generations? The paper assumes it is self-evident that we should not violate human rights as described in the UDHR, but what do rights mean to future generations that do not currently exist? Firstly, it would seem that global justice's whole endeavor is to improve the future; scholars identify worldwide injustices in the present and explain why it is wrong for different entities to contribute to these injustices, then people read their papers in the future. This argument is insufficient to explain why people in the present may have to give up some luxuries so that people in the future live better. However, it seems like bettering the lives of future generations is entailed by any global justice argument.

Premise Three:

The third premise is that countries have obligations to future generations. As we have described, climate change threatens access to food, water, and health, thus the lives of millions of people. In the third chapter of his book *Climate Change, Justice and Future Generations*, Edward Page explains how climate change poses an issue for a theory of justice called "basic capability egalitarianism"(Page 67). To prove the thesis and show why current generations have obligations to future generations, a theory of justice seems most appropriate to explain

intergenerational responsibilities fully. Page's descriptions provide an excellent account of why equality of capabilities is an effectively logical theory to address harms to future generations.

Thus far, the paper has gone into detail about extreme climate events that affect people's abilities to address their basic needs in the present. We have briefly explained how climate change will affect the future but have not gone into any specifics. We will need these specifics to describe why countries have obligations towards future generations adequately. According to the "IPCC"(Page 70), climate change will hamper the ability of "global agriculture"(Page 71) to meet the needs of the future human population. We should keep in mind that at the moment, around "640" million people are "at risk of hunger"(Page 71). By "the year 2060", there could be "300"(Page 70) million more people who are food insecure on top of the 640 million who are already food insecure. There will also be extensive flooding due to "sea-level rises" that will cause "infertility"(Page 70) in the soils along coasts and leave many areas uninhabitable. This is all to say, the damage that climate change will cause in the future will probably be orders of magnitude worse than what it has already done in the past. Scientists estimate that mitigation costs could reach over a trillion a year "by 2100"(Ackerman) for the US alone. As we will see, these facts are extremely problematic for basic capability egalitarianism.

This kind of distributive egalitarianism maintains that we should seek an equitable distribution of "basic capabilities"(Page 70) so that people need to live a life worth living. Page outlines that according to "Martha Nussbaum"(Page 67), these capabilities include but are not limited to "1. Life; 2. Bodily Health; 3. Bodily Integrity," and "10. Control over one's environment"(Page 69). From this description, we can see that "climate change is a clear and present threat to"(Page 71) to the basic capabilities of people in the future. This theory of justice is appealing because it does not seek to acquire absolute equality; it recognizes that, for example,

a smaller person with a lower metabolism would probably need less food to be “well-nourished than a person with a higher metabolic rate”(Page 68). The observation that everyone is intrinsically different, and will not need the same things to “achieve valuable functionings”(Page 68), coupled with the fact that this theory only focuses on the basics, makes it much more difficult to argue against it. For a country like Norway, which has a robust social safety net, this theory will not find too many injustices. It is difficult to say that a wealthy Norwegian should equalize themselves with a poorer one because the poorer one will always be food secure. However, it is easier to argue that both Norwegians should equalize themselves with a food insecure person in the developing world along these egalitarian lines.

What does basic capability theory mean intergenerationally? We know to a high degree of certainty that people will continue to have children, and these children will have the right to these basic capabilities as soon as they are born. So there can still be an expected harm to these expected people. These expected people “have a *prima facie* claim at the bar of distributive justice”(Page 68), as much as anyone living today does. This egalitarianism requires that:

“present persons, and the states they belong to, should not act so as to undermine the possibility that members of future generations will enjoy their basic capabilities to function”(Page 70)

This distributive egalitarianism includes every person; everyone has the right to access these essential capabilities. We can reasonably assume that climate change will threaten the basic capabilities of millions of future people. So if countries have a way to allot these capabilities to future generations without hindering people today, they are obligated to do what they can do

allot the capabilities. Governments may have to hold off on huge infrastructure spending, upgrading school buildings, building new airports, or anything that does not hinder their ability to provide the basics to their current citizens to mitigate the effects of climate change. This kind of action is necessary to ensure that future humans can live comfortably. We should note that this theory also applies to a state's internal and "international"(Page 70) relations in the present, so it bolsters the argument the paper has made for why countries need to start mitigating climate change right now to secure the needs of people in the present. The paper will therefore adopt this egalitarian logic to explain obligations to future generations.

Both cluelessness and the non-identity problem pose significant epistemic problems for basic capability egalitarianism. However, the arguments the paper has already made to refute these objections pertain to this theory of justice. When talking about countries and their emissions, we are not scientifically clueless, and Hiller's argument that we have *prima facie* responsibilities applies. In terms of the non-identity problem, we must simply take Caney's view that even if we cannot say that individuals have been adversely affected by climate change, we can certainly say that groups or countries have been. So if we look at future people as a group, or the people within future countries as a bunch of groups, we can see that climate change will hinder the group's abilities to achieve their basic capabilities.

This argument suggests that we should not hinder the basic capabilities of people alive today to mitigate climate change. Developing countries should prioritize the basic needs of their citizens today before they consider people who do not yet exist. Countries must reduce their emissions as much as possible while also improving their current populations' material well-being. Maybe a country should not need to upgrade its schools to use the latest iPads, but if an area has no schools, they should probably build some.

Now one may ask - what if every country does or can not take up this issue? Can the world fix climate change if every country does not stop its emissions? Again, scientists suggest that any significant reduction in fossil fuel emissions that helps get us below the average of 1.5 degrees Celsius of warming would be worth pursuing(Broom). For example, If the US, which contributes around 15% of yearly emissions, somehow cut all of its emissions, it would reduce global emissions by almost a sixth(UCSUSA). This scenario is hyperbolic, but it is just to show that every country does not need to reduce their emissions equally to make a difference. Any government reducing its emissions as much as possible will help itself and every country. It elongates the time that scientists have to perfect alternatives to fossil fuels.

Conclusion:

By burning fossil fuels, humans emit GHGs that cause climate change, which increases both the frequency and severity of extreme weather events. The argument for premise one explains that even if using fossil fuels is not the same as a crime like murder, it still causes measurable harm to people, so there is a *prima facie* duty for states to emit as little as possible. In advocating for premise two, the paper outlines that the benefits countries get from their current emissions ground an even stronger obligation to take drastic action to mitigate their emissions. The third and last premise uses the egalitarian perspective that everyone needs access to basic capabilities to live a life worth living to argue that countries need to consider people's abilities in the future and the present. This consideration means that governments need to take drastic action to mitigate climate change and safeguard people's basic capabilities in the future. The paper hopes that this argument motivates people to help their countries fulfill the obligations outlined in the and take the drastic actions they will need to take to save our planet. There is still hope that

we can avoid becoming Venus, but we need to innovate and make convincing arguments for people who are skeptical.

Reflection:

Firstly, I must state that throughout my time at Northeastern, this capstone has been the most challenging philosophy paper I have ever had to write, which makes sense because this is a capstone. The class and the paper have incredibly high standards. This means that I have learned more in writing this paper than I have writing any other philosophy paper. I tried my best and I am happy with the finished product. I(as many other young people) am incredibly passionate about climate change mitigation, so it was a pleasure to write the paper. I am glad that I was able to have my work scrutinized so harshly and frequently because it forced me to try to make the most convincing arguments possible.

This paper was especially difficult for me because I had not yet taken this kind of applied philosophy class before. My degree is Computer Science and Philosophy, so my past classes focused more on things like sentential logic, or making basic arguments where you have premises that if true lead to a conclusion. It was not as easy to have a false premise in my other classes. Further, I had never needed to ensure that I engaged with the literature on a topic to the extent I had to in this paper, and I found it difficult to present all the relevant arguments(climate change is a huge topic), whilst also making my own arguments. However, looking at the way other philosophers cited each other made this easier. I also found it difficult to tow the line between philosophical argumentation and practical applications, I was unsure how many facts I needed to present for my arguments, whilst also making sure I was not writing a political science paper. Due to all of this, I probably wrote over 10,000 words for this paper, and I did not use most of what I wrote, I had to scrap most of my first draft because I was not making enough arguments, just summarizing other philosophers, but a lot of what I wrote at first was useful to

understand the other philosopher's arguments. What I had previously written, plus the scrutiny I got made it a lot easier to write my final draft because I knew exactly how I wanted to make my argument, I just had to write it down.

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